

AtlasPredict

Project Team

Umar Gul 20I-0591
Omar Mughal 20I-0739
Zainab Kayani 20F-1134

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Supervised by

Dr. Atif Aftab Ahmed Jilani

Co-Supervised by

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Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

This document presents the initial development phase of AtlasPredict, a real-time disaster prediction system designed specifically for Pakistan. AtlasPredict aims to predict natural disasters such as floods, heavy rainfall, and landslides by integrating geospatial data from authoritative sources such as NASA and NDMA, and applying advanced machine learning models. This system is intended to provide actionable insights through a user-friendly platform, improving disaster preparedness and response. This report outlines the system's problem statement, project scope, and key modules, along with a detailed description of the target users and their characteristics.

The report is primarily aimed at developers and technical teams who will be involved in designing and implementing the solution, project managers overseeing development progress, and academic supervisors for evaluation purposes. Secondary audiences include potential users such as government agencies, emergency responders, and disaster management teams who will interact with the final system to receive real-time disaster alerts and decision support tools.

1.1 Problem Statement

Natural disasters, particularly floods, heavy rainfall, and landslides, pose significant risks to Pakistan, a country highly susceptible to these events due to its diverse geography and climate. Despite the urgent need for effective disaster preparedness and mitigation, existing disaster management systems in Pakistan are largely reactive and lack the precision required for timely interventions. Current systems do not fully utilize the available geospatial data from global and local sources such as NASA, NDMA, SUPARCO, and PMD, resulting in fragmented analyses and insufficient early warnings. As a result, emergency response teams, government agencies, and affected communities are often ill-prepared to handle the imminent risks, leading to unnecessary casualties, property damage, and

extended recovery periods.

AtlasPredict addresses this critical gap by providing an advanced machine-learning-powered disaster prediction system that leverages diverse geospatial datasets to offer precise and real-time disaster forecasts. By providing accurate, timely warnings and localized risk analysis, the system aims to enhance disaster preparedness, allowing authorities and communities to respond proactively and effectively. AtlasPredict aspires to bridge the technological gap in Pakistan's disaster management framework, reducing human and economic losses and fostering a more resilient response strategy.

1.2 Scope

The scope of AtlasPredict involves the development of a comprehensive disaster prediction platform tailored for Pakistan's unique disaster landscape. The system will integrate multi-source geospatial data, including satellite imagery and environmental datasets from NASA and NDMA, with the goal to also include SUPARCO and PMD, to offer highly localized predictions for floods, heavy rainfall, and landslides. Advanced machine learning models such as Random Forest and CNN will be employed for initial flood predictions, while future iterations will incorporate LSTM and SVM models to enhance prediction accuracy over time.

Key functionalities include real-time data streaming, automated alerts, and an interactive web-based platform featuring the Pakistan Atlas, where users can visualize disaster-prone areas and weather forecasts. The system is designed to be scalable, allowing for future expansion to other disaster types such as earthquakes and droughts. Additionally, the project will incorporate features such as community reporting, allowing local users to input real-time data to further enhance prediction accuracy. The platform will be optimized for user accessibility, with the ability to provide real-time, region-specific alerts in multiple formats (e.g., text, web notifications). The boundaries of the project include data collection, prediction model development, and the deployment of a fully functional platform, excluding the physical deployment of emergency response resources.

In summary, AtlasPredict's primary objective is to provide a centralized, real-time disaster prediction platform that improves disaster preparedness and response in Pakistan, leveraging cutting-edge technologies and scalable infrastructure.

1.3 Modules

AtlasPredict consists of several key modules that work together to offer accurate predictions and provide users with actionable insights. The modules are divided into client-

facing and backend components to ensure seamless operation and user interaction.

1.3.1 Client Web Application

This module serves as the main interface for end-users such as government authorities, emergency responders, and the public, allowing them to interact with real-time predictions, receive alerts, and access disaster data.

1. **User Dashboard:** Displays disaster predictions, risk levels, and localized alerts in an intuitive, user-friendly interface.
2. **Interactive Pakistan Atlas:** Provides users with dynamic geospatial visualizations, allowing them to explore disaster-prone areas such as flood zones in real time.
3. **Alert Notification System:** Sends region-specific, real-time alerts for impending disasters, enabling timely response actions.

1.3.2 Admin Web Application

This module is accessible by system administrators responsible for data integrity, model training, and system performance monitoring.

1. **Data Management Interface:** Facilitates the integration, validation, and monitoring of datasets from NDMA and NASA.
2. **Machine Learning Model Management:** Enables administrators to update and fine-tune prediction models based on incoming datasets.
3. **System Monitoring:** Tracks the overall performance of the prediction system, ensuring data flows and model accuracy.

1.3.3 Machine Learning and Prediction Engine

Following are the requirements for this module:

1. **Data Preprocessing:** Ensures incoming datasets from sources like NASA and NDMA are cleaned and normalized for machine learning analysis.
2. **Predictive Modeling:** Uses machine learning algorithms (Random Forest, CNN) to predict natural disasters such as floods and heavy rainfall.

3. **Model Retraining:** Automatically retrains the model when new data is integrated to improve prediction accuracy.
4. **Real-Time Data Integration:** Continuously ingests and updates the system with real-time weather and satellite data for up-to-date predictions.

1.4 User Classes and Characteristics

AtlasPredict is designed to serve multiple types of users, each with distinct roles and interaction needs. The primary users include government disaster management authorities, emergency responders, and the general public in disaster-prone areas. Additionally, system administrators are responsible for maintaining the system's accuracy and integrity. The following table outlines the key user classes and their characteristics:

User Class	Description
Government Authorities	This class includes national and provincial disaster management authorities such as NDMA and PDMA . They require real-time data and accurate forecasts to make critical decisions about resource allocation, evacuation strategies, and response coordination. These users interact with the platform via the admin dashboard to monitor disaster predictions and access region-specific alerts.
Emergency Responders	Organizations like Rescue 1122 and local fire, medical, and rescue teams. These users need real-time alerts and predictive data to mobilize quickly in response to disasters. They primarily use the alert notification system and the mobile-friendly web interface for critical updates.
General Public	Citizens in disaster-prone areas such as floodplain communities or mountainous regions at risk of landslides. These users access region-specific alerts through notifications (SMS and web) and visualizations on the Pakistan Atlas. They may also contribute to data via the community reporting feature .
System Administrators	Technically proficient users responsible for ensuring data integrity and managing the machine learning models. They use the admin web application to integrate datasets, validate data, and manage model training, ensuring the accuracy and scalability of the predictions.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter outlines the functional and non-functional requirements for AtlasPredict, a real-time natural disaster prediction system for Pakistan. The system integrates geospatial data and machine learning models to provide real-time predictions and alerts. The requirements are divided into use cases, functional requirements, and non-functional requirements, each focusing on the specific needs of the system and its users.

2.1 Use-case/Event Response Table/Storyboarding

AtlasPredict is a real-time system, where most functionalities are handled by backend processes such as data ingestion, prediction model execution, and alert generation. An Event-Response Table is therefore the most suitable method for capturing its behavior.

Event	System Response
New satellite/weather data received	System ingests the data from external sources (e.g., NASA, NDMA, SUPARCO) and preprocesses it for analysis.
Geospatial data anomalies detected	The machine learning model identifies potential disaster-prone areas based on anomalies (e.g., sudden rainfall increase, water level rise). Risk is classified.
High-risk flood area detected	An alert is generated and sent to authorities, responders, and users. The map is updated in real-time.
User requests flood prediction	The system retrieves the latest model predictions and displays the risk classification and advisories on the dashboard.
Community-reported data received	The system validates crowdsourced data, incorporates it into prediction models, and refines forecasts accordingly.
Model accuracy threshold crossed	The system initiates model retraining using new data to improve prediction accuracy.
End-user accesses alert history	The system retrieves and displays past disaster alerts and predictions for review.

Table 2.1: Event-Response Table for AtlasPredict

2.2 Functional Requirements

The functional requirements of AtlasPredict are grouped into 3 key modules: the Client Web Application, Admin Web Application and Machine Learning and Prediction Engine:

2.2.1 Module 1: Client Web Application

The client web application serves as the main interface for government authorities, emergency responders, and the general public. The functional requirements for this module are:

1. FR1: The web application shall display real-time disaster predictions (floods, heavy rainfall) based on satellite and weather data.
2. FR2: The system shall provide users with localized, customizable alerts.

3. FR3: The system shall allow users to access the interactive Pakistan Atlas for real-time data visualizations of high-risk areas.
4. FR4: Users shall receive region-specific advisories, including evacuation instructions based on risk classification.
5. FR5: The system shall allow users to review past predictions and advisories via an alert history.

2.2.2 Module 2: Admin Web Application

The admin web application is used by system administrators to manage data, update machine learning models, and monitor system performance. The functional requirements for this module are:

1. FR1: The admin interface shall allow administrators to validate and integrate datasets from sources such as NASA and NDMA.
2. FR2: The system shall enable administrators to retrain machine learning models based on new data inputs.
3. FR3: The system shall allow administrators to monitor system performance, data integrity, and prediction accuracy in real-time.
4. FR4: Administrators shall be able to manually trigger alerts or override automated predictions when necessary.
5. FR5: The admin web application shall support multi-factor authentication for secure access.

2.2.3 Module 3: Machine Learning and Prediction Engine

The Machine Learning and Prediction Engine is responsible for processing data and generating disaster predictions. The functional requirements for this module are:

1. FR1: The system shall ingest geospatial datasets (satellite imagery, weather data) from NASA and NDMA, and preprocess the data for analysis.
2. FR2: The machine learning models (Random Forest, CNN) shall generate predictions of floods and heavy rainfall.
3. FR3: The system shall automatically retrain models based on new data for improved accuracy.

4. FR4: The prediction engine shall trigger alerts when the risk threshold for a disaster is crossed.
5. FR5: The system shall validate and incorporate community-reported data in real-time to refine predictions.

2.3 Non-Functional Requirements

The non-functional requirements outline the quality and performance expectations of the system to ensure reliability, usability, and security.

2.3.1 Reliability

- The system shall maintain an availability of 99.9% with an MTBF (Mean Time Between Failures) of 12 months.
- Errors in data ingestion or model execution shall trigger alerts in the admin dashboard, and any prediction errors shall lead to automatic model retraining.

2.3.2 Usability

- USE-1: The system's user interface shall allow users to navigate to real-time disaster predictions and alerts within three interactions.
- USE-2: The interface shall be mobile-compatible, allowing users to access predictions on smartphones and tablets.

2.3.3 Performance

- PER-1: The system shall process incoming satellite data within 10-30 seconds, updating predictions in real-time.
- PER-2: The client web application shall display prediction results and visualizations within 5-10 seconds on a standard internet connection.

2.3.4 Security

- SEC-1: All data transmissions between the web application and backend servers shall be encrypted using SSL/TLS.

- SEC-2: Multi-factor authentication (MFA) shall be required for administrators accessing the data management and machine learning modules.

2.4 Domain Model

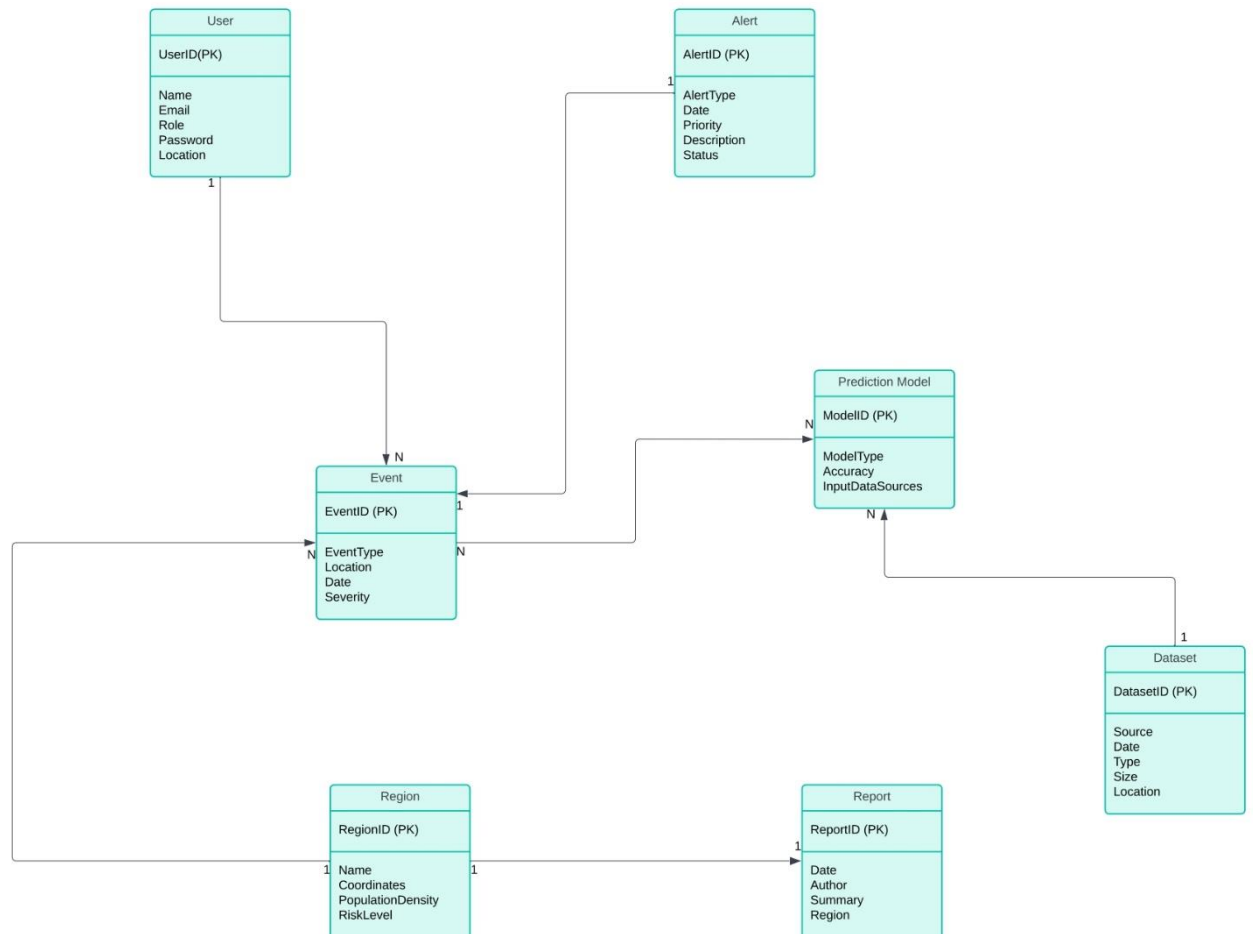


Figure 2.1: AtlasPredict Domain Model

Chapter 3

System Overview

General Description of AtlasPredict

AtlasPredict is a natural disaster prediction system designed to analyze geospatial and environmental data for early warning and risk assessment. The core functionality of the system is to process large datasets, such as those provided by NASA, and apply machine learning algorithms to predict potential natural disasters like floods, earthquakes, and extreme weather events. The system is tailored to provide insights for regions vulnerable to natural disasters, helping governments, emergency services, and communities to prepare and respond proactively.

Context

Natural disasters pose a significant threat to lives, infrastructure, and economies. Accurate and timely prediction of such events is critical for reducing their impact. AtlasPredict is developed within this context, aiming to bridge the gap between available geospatial data and actionable insights. The system is focused on countries like Pakistan, where the need for disaster management is acute due to frequent natural calamities. By utilizing satellite data, climate trends, and historical disaster records, AtlasPredict provides valuable predictions to aid in disaster preparedness efforts.

Design

The design of AtlasPredict revolves around three key components: 1. Data Collection and Processing: Geospatial data from NASA and other sources is collected, cleaned, and structured for analysis. This data includes climate patterns, topography, historical disaster data, and environmental factors.

2. Machine Learning Models: Predictive models are at the heart of the system, trained to identify potential disaster risks based on past data and environmental conditions. These models continuously learn from new data to enhance prediction accuracy over time.

3. User Interface: The system features a user-friendly interface that allows users to in-

interact with the predictions and view the risk assessments. The dashboard displays maps, graphs, and other visualization tools to communicate the analysis in an intuitive way.

Background Information

AtlasPredict builds upon research in geospatial data analysis, machine learning, and disaster management. The project leverages technologies like PostGIS for spatial data handling, Python-based machine learning frameworks, and cloud storage solutions for efficient data management. Its design and functionality are inspired by the growing need for intelligent disaster risk reduction systems that can harness the power of big data and machine learning to predict natural events and help mitigate their consequences.

3.1 Architectural Design



Team Members

Umar Gul 20I-0591
Omar Mughal 20I-0739
Zainab Kayani 20F-1134

Supervisor

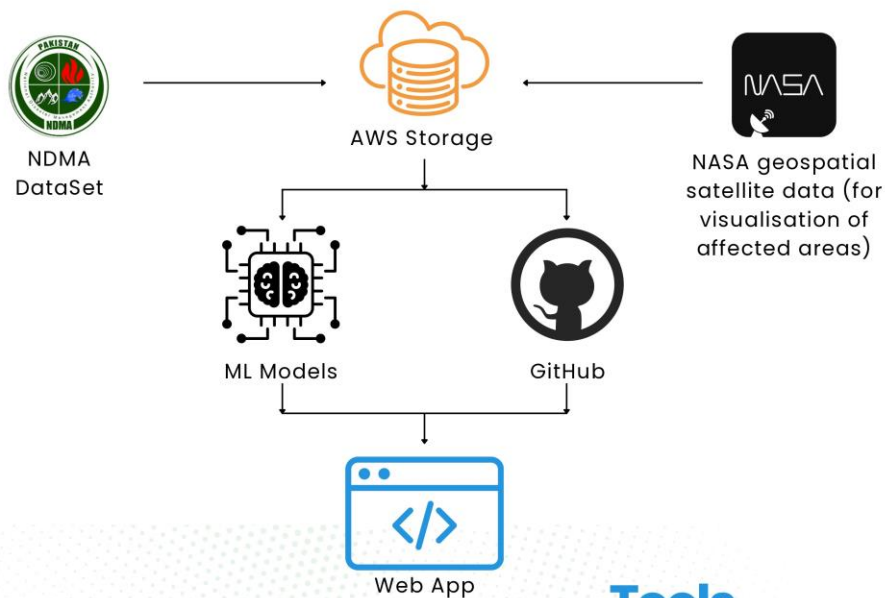
Dr. Atif Aftab Ahmed Jilani



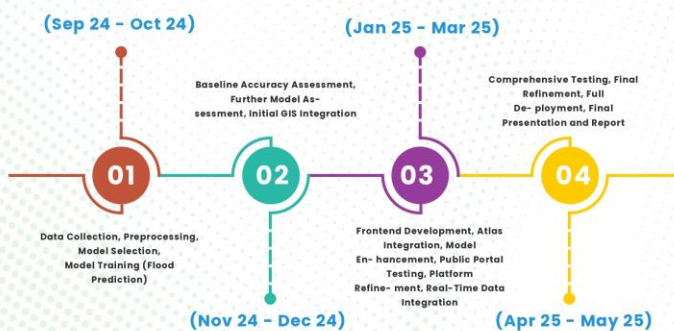
Objectives

Develop a disaster prediction system for Pakistan using advanced machine learning and diverse datasets. Provide early warnings to stakeholders for better disaster preparedness. Aim to reduce casualties, economic losses, and recovery times.

Architecture Diagram



Timeline



Tools



Figure 3.1: AtlasPredict Architectural Diagram

1. Data Sources

NASA and NDMA: Data is collected from two main sources, NASA (satellite-related data) and NDMA (disaster management data).

2. Data Storage

AWS Storage: The collected data is sent to AWS storage (e.g., S3) for secure storage and easy access.

3. Processing Pipelines

ML Models: The data from AWS storage is used by machine learning models to perform analysis or predictions.

GitHub: Simultaneously, the data or code related to the models might be pushed or version-controlled on GitHub.

4. Data Processing

Processing: After the ML models are run, the data is processed, cleaned, and prepared for presentation. Random Forest and CNN models for initial flood prediction; LSTM and SVM models for future enhancements.

5. Output

Web App: Finally, the processed data is sent to a web application for visualization or user interaction.

3.2 Design Models

Activity Diagram:

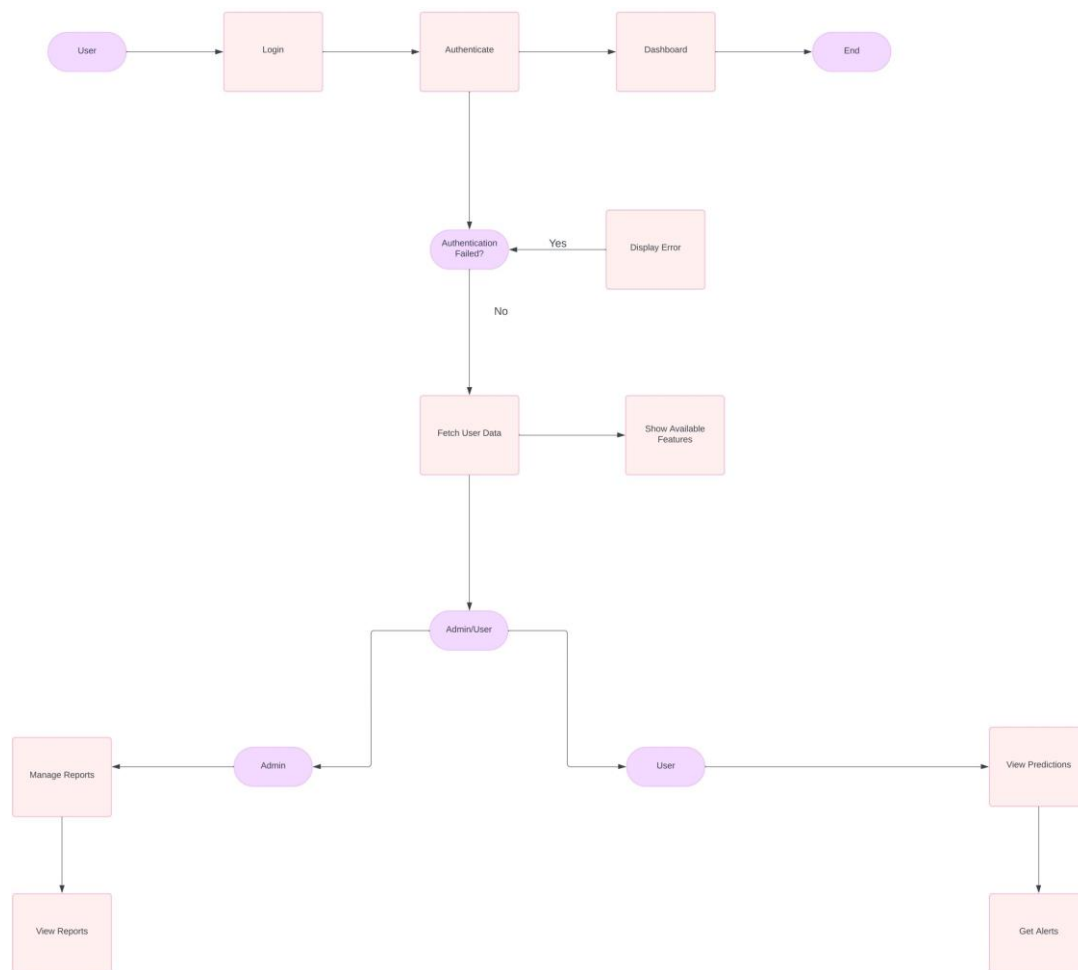


Figure 3.2: AtlasPredict Activity Diagram

The activity diagram shows the flow of control and the sequence of actions in the system. It is useful to describe the dynamic behavior of the system and how different subsystems interact.

Key Activities in AtlasPredict:

- **User Login and Authentication**
- **Data Acquisition from External Sources**
- **Prediction Model Execution**
- **Alert Generation Based on Prediction**
- **Event and Region Monitoring**
- **Report Generation**

Class Diagram:

Figure 3.3 shows the class diagram for AtlasPredict on next page.

Key Classes in AtlasPredict:

- **User:** Attributes like UserID, Name, Role, Email. Methods like login(), viewDashboard(), etc.
- **Event:** Attributes like EventID, EventType, Location. Methods like getDetails(), getSeverity().
- **PredictionModel:** Attributes like ModelID, Type, Accuracy. Methods like runModel(), generatePrediction().
- **Dataset:** Attributes like DatasetID, Source, Type. Methods like fetchData().
- **Alert:** Attributes like AlertID, Priority, Description. Methods like sendAlert().
- **Region:** Attributes like RegionID, Coordinates, RiskLevel. Methods like getRisk().
- **Report:** Attributes like ReportID, Summary, Date. Methods like generateReport().

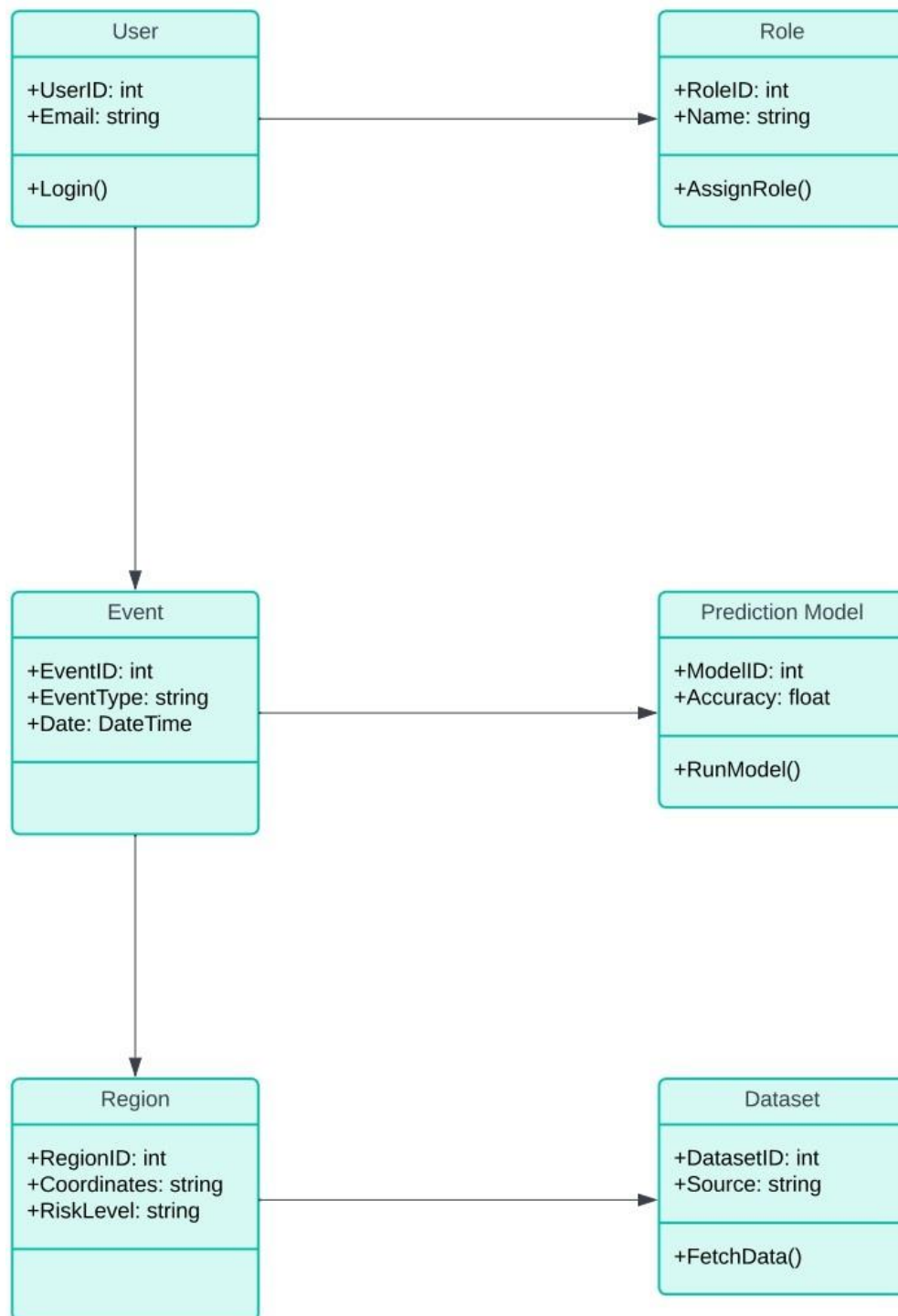
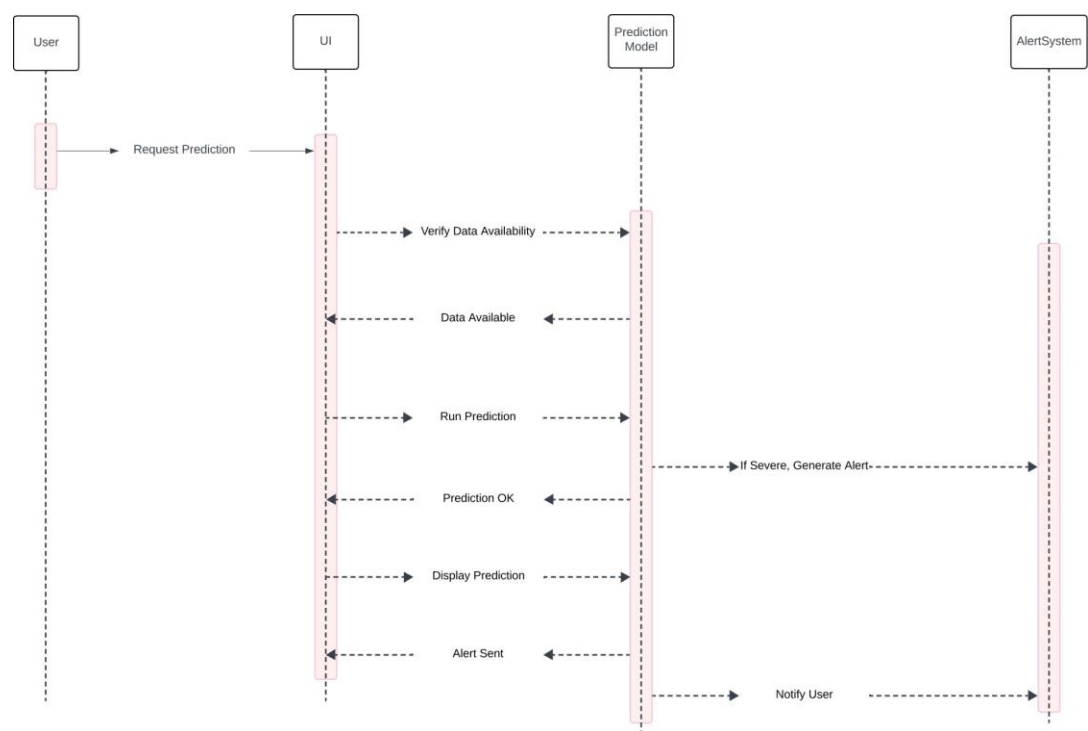


Figure 3.3: AtlastPredict Class Diagram

Class Level Sequence Diagram:
Example Scenario: Prediction and Alert Generation

Figure 3.4: AtlasPredict Class Level Sequence Diagram



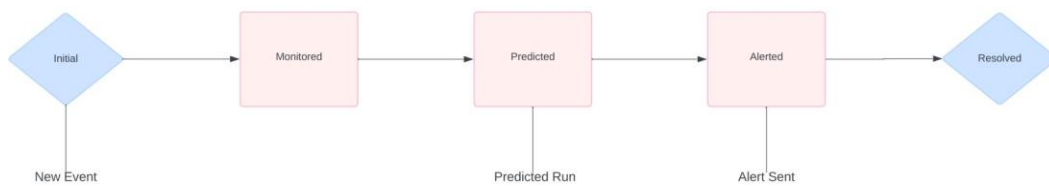
Explanation:

- **User** requests a prediction through the **UI**.
- The **UI** checks if relevant data is available in the **PredictionModel**.
- The **PredictionModel** runs the prediction and checks for severity.
- If severe, the **AlertSystem** is triggered to generate an alert.
- The **UI** displays both the prediction and the alert (if generated) to the user.

State Transition Diagram:

This diagram shows the various states an object (or the system) can be in and how it transitions between these states in response to different events.

Figure 3.5: AtlasPredict State Transition Diagram



Example: Event Lifecycle in AtlasPredict

- **States:** Initial, Monitored, Predicted, Alerted, Resolved
- **Transitions:** Triggered by actions such as a new event being logged, prediction generation, or alert dispatch.

3.2 Data Design

For AtlasPredict, the information domain can be transformed into data structures by breaking it down into key entities related to natural disaster prediction and geospatial analysis. Here's an overview of how the major data/system entities can be structured and organized:

1. Data Structures

- **Geospatial Data (NASA Datasets):** This will include multi-dimensional arrays or matrices to store data points representing geographical locations (latitude, longitude, altitude), along with time-series data capturing changes over time. These data structures can be optimized using geospatial libraries like GeoPandas or GDAL.
- **Disaster Risk Models:** Based on machine learning algorithms, this data might involve data structures like decision trees, neural networks (for deep learning models), or regression models. These structures would process inputs like weather patterns, historical data, and other environmental factors.
- **Event Logs:** Information regarding historical disasters (earthquakes, floods, etc.) can be stored in structured formats like hash maps, arrays, or tables to allow quick lookup and reference. This can be stored in an SQL database.
- **User Data:** User-related information, such as users accessing the system, can be organized into classes or dictionaries representing different attributes like name, email, user role, and access logs.

2. Entity Storage and Processing

- **Geospatial Data:** This data could be stored in specialized geospatial databases like PostGIS, which extends PostgreSQL to handle geospatial queries. It allows efficient querying, indexing, and manipulation of geographical data. Alternatively, cloud storage solutions like AWS S3 could be used for larger datasets.
- **Machine Learning Models:** These models can be stored as serialized objects (using

formats like Pickle in Python or ONNX for machine learning models), and loaded into memory for processing when needed. Cloud-based model serving (e.g., using AWS SageMaker) could be an option for large-scale deployment.

- **Prediction Results:** Predictions and analysis results could be stored in relational databases like PostgreSQL or NoSQL databases like MongoDB for rapid access and updating, especially when handling unstructured data like user queries and system logs.
- **Historical Data:** Information on past natural disasters and patterns can be stored in relational databases like MySQL or PostgreSQL, allowing for easy querying and processing to train the machine learning models or for generating insights.

3. Databases/Data Storage Items

- **Primary Database:** PostgreSQL (with PostGIS extension) or MongoDB for flexible storage of both structured and unstructured data.
- **Geospatial Data Storage:** AWS S3 or Google Cloud Storage for raw geospatial data.
- **Machine Learning Model Storage:** Pickle/ONNX for serialized models, with cloud-based storage options like AWS S3 or Azure Blob.
- **User/Access Data:** Stored in relational databases like MySQL or SQLite for user-related information.
- **Event Log Storage:** Could be managed through a logging framework with storage in SQL databases or NoSQL databases like MongoDB.

By organizing and processing data in this way, AtlasPredict can efficiently handle large datasets, geospatial information, and machine learning models to provide accurate disaster prediction and insights.

Bibliography